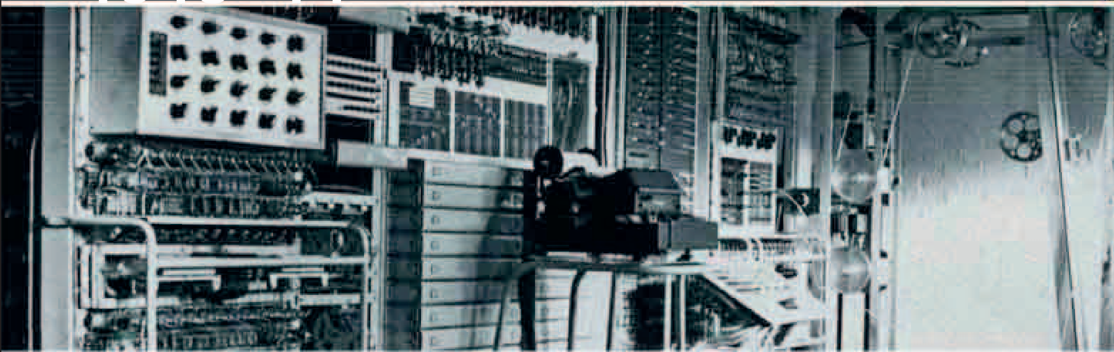
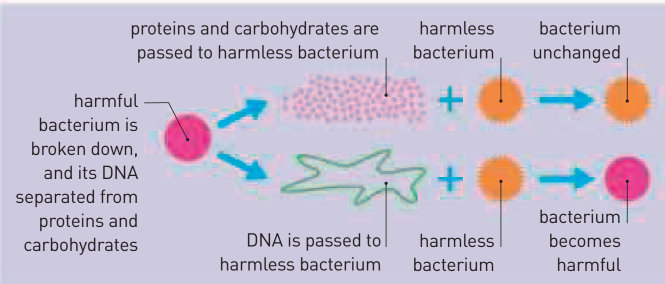
1943–44



These are the control panels of the first electronic programmable computer—Colossus—which was used to decode German messages during World War II.



DISCOVERING THE NATURE OF DNA

Oswald Avery sought to identify the chemical substance that could transform harmless bacteria into harmful bacteria. Avery killed harmful bacteria and broke them down into their various chemical components (protein and carbohydrates and DNA). He added each component in turn to harmless bacteria, and he found that only the bacteria's DNA could cause a change.

DURING WORLD WAR II, the pioneering code-breaking work of British mathematician **Alan Turing** (1912–54) triggered an outburst of computer technology. One of the **earliest electronic digital computers** was built in Britain in 1943: **Colossus**.

French engineer **Émile Gagnan** (1900–79) modified a gas-regulator valve for a new use. Working with French biologist **Jacques Cousteau** (1910–97), he adapted the device so that it could be used to control the air supply in an aqualung. This invention would **revolutionize underwater exploration**.

A major step forward in unlocking the science of genes and inheritance was made in 1944. Canadian-born physician **Oswald Avery** (1877–79) wanted to identify the transforming principle first identified by Frederick Griffith (see 1928). Avery performed similar experiments to Griffith, but he analyzed the genetic factor in more detail. He demonstrated that if proteins and their enzymes were removed from bacteria, another chemical present—DNA—could still cause transformation. This showed that **genes are made of DNA**, not protein, as was previously thought.

Quinine, a naturally occurring substance found in the the bark of the South American cinchona tree, had long been valued for its antimalarial properties. Supply had become difficult during World War II, but in May 1944, American chemists **Robert Woodward** (1917–79) and **William Doering** (1917–2011) announced that they had successfully manufactured it.

Austrian physician **Hans Asperger** (1906–80) had been studying mental disorders in children and **formalized the diagnosis of autism**. He examined a group of autistic children who, because of the nature of their way of learning, he called little professors. Their condition later became known as **Asperger's syndrome**.

> " WHY DIDN'T **AVERY** GET THE **NOBEL PRIZE?** BECAUSE MOST **PEOPLE DIDN'T TAKE HIM SERIOUSLY.** "
>
> **James Watson, American geneticist,** from *Nature*, April 1983

March 1943 Construction of **Colossus computer** begins at Bletchley Park, UK

April 1, 1943 A system that **encrypts transatlantic telephone signals** (SIGSALY) is launched

Summer 1943 Jacques Cousteau tests new **compressed-air aqualung**

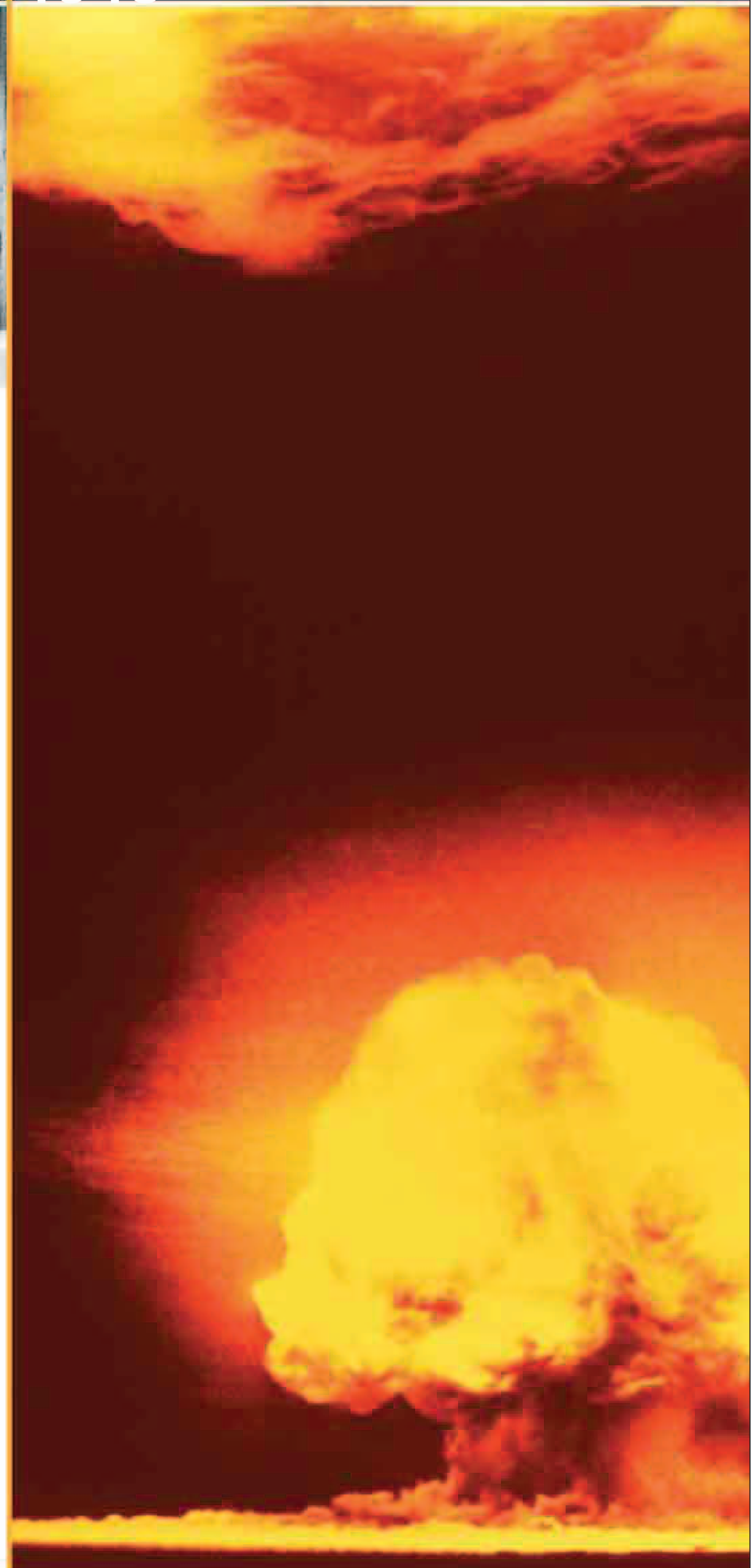
February 1, 1944 Oswald Avery and Canadian and American scientists **Colin MacLeod** and **Maclyn McCarty** confirm that genes are made from DNA

May 1944 Robert Woodward and William Doering describe the **synthesis of quinine**

June 3, 1944 Hans Asperger describes a **form of autism** that would later become known as **Asperger's syndrome**

December 1944 American zoologist **Donald Griffin** describes the orientation behavior of bats and coins the term **echolocation**

1945



February Dorothy Hodgkin describes the central chemical **structure of penicillin**

February 16 American physician Raymond L. Libby describes a method of delivering **penicillin by mouth**

March American chemists **Jacob Marinsky, Lawrence Glendenin**, and **Charles Coryell** isolate a new element—**romethium**